# Diego Canales

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## RESEARCH INTERESTS

Primary: Political Economy, Game Theory, Electoral Competition. Secondary: Applied Microeconomics, Economic History.

## EDUCATION

**PhD Economics** **2021**
University of Nottingham, School of Economics, United Kingdom

**M.Sc. Mathematics** **2014**
University of Texas at El Paso

**B.Sc. Applied Mathematics** **2011**
University of Texas at El Paso

**M.A. Economics** **2014**
Universidad Autónoma de Ciudad Juárez

**B.A. International Business** **2013**
Tecnológico de Monterrey, Campus Ciudad Juárez

## PUBLISHED JOURNAL PAPERS

**A Spatial Model with k-winners.** ***Games*** **2025, 16(1), 6.**

The concept of the Condorcet winner has become central to most electoral models in the political economy literature. A Condorcet winner is the alternative preferred by a plurality in every pairwise competition; the notion of a k-winner generalizes that of a Condorcet winner. The k-winner is the unique alternative top-ranked by the plurality in every competition comprising exactly k alternatives (including itself). This study uses a spatial voting setting to characterize this theoretical concept, showing that if a k-winner exists for some k>2, then the same alternative must be the k'-winner for every k'>k. We derive additional results, including sufficient and necessary conditions for the existence of a k-winner for some k>2.

## TEACHING / ADMIN EXPERIENCE

**Assistant Professor of Economics** **06/2026 – Present**
Tecnológico de Monterrey – Economics Dept., Campus Puebla

**Full-Time Professor** **07/2022 – 06/2026**
Tecnológico de Monterrey – Economics Dept., Campus Puebla

**Teaching Associate/Fellow** **09/2019 – 08/2022**
The University of Nottingham, School of Economics

**UG/PG Dissertation Supervisor** **10/2019 – 08/2022**
The University of Nottingham, School of Economics

**Graduate Teaching Assistant** **10/2016 – 08/2019**
The University of Nottingham, School of Economics and Business School

**Mathematics Lecturer** **08/2015 – 12/2015**
The University of Texas at El Paso, Math Department

## WORKING PAPERS & UNDER REVIEW

**Electoral Competition and the Design of Tax and Spending Policies in Sequential Elections**

This paper develops a political economy model analyzing how primary election formats, open versus closed, affect both the ideological moderation and productive efficiency of fiscal policy. We characterize subgame perfect Nash equilibria in a sequential election game where voters first nominate candidates in party primaries, then elect officials in a general election. Our main contribution is identifying sufficient conditions under which (1) open primaries produce moderate but inefficient government spending, while (2) closed primaries produce efficient but polarized fiscal policy. These results reveal a previously unrecognized trade-off in electoral system design.

**The Case of American Presidential Primaries: A Model of Sequential Primary Elections**

We present a theoretical model to analyse two phenomena surrounding sequential elections: momentum and Duverger's law. Momentum can be defined when a candidate starts the election season with a streak of successful events (polling or election results) in such a way that it would guarantee a greater probability of winning the race. Our modified version of Duverger's law asserts that primary candidates tend to quit at earlier stages of the election under plurality rule compared to proportional representation. We exploit the dynamic nature of primary elections using the concept of a Condorcet-winner and k-winners.

**Subnational Heterogeneity in Maternal Employment Recovery from Child Penalties: Descriptive Evidence from Mexico**

Documents child penalties on maternal employment across Mexico's 32 states using a pseudo-event study on pooled cross-sections from the ENIGH (2016–2024, N≈450,000 parents). The initial mother–father employment gap averages ~52 percentage points; the state-level standard deviation of that gap is 4.2 pp, while the standard deviation of maternal recovery as children age is larger in both absolute and relative terms. The association between local informality and maternal employment grows monotonically with child age, with no analogous gradient for fathers, and the informality–age profile loads entirely on informal jobs. Findings are presented descriptively, not causally.

**Within-Person Drivers of BMI Change in Aging Mexico**

Fixed-effects panel analysis showing that the cross-sectional wealth–BMI gradient largely reflects between-person heterogeneity rather than a within-person causal effect, with depression and multimorbidity emerging as the main health drivers. Data: MHAS/ENASEM Harmonized 2001–2021 (~18,000 respondents). Presented as a poster at the International Congress on Obesity Research (IOR) 2026, Monterrey; manuscript in preparation.

**Treaty Citizens and Immigrants: Intergenerational Mobility Among Mexican-Origin Populations in the American Southwest, 1850–1940**

The Treaty of Guadalupe Hidalgo (1848) granted U.S. citizenship overnight to approximately 80,000 Mexican nationals residing in the ceded territories, the largest such grant in U.S. history. Using the IPUMS Multigenerational Longitudinal Panel, this paper follows sons of "treaty citizens," sons of Mexican immigrants, and sons of white natives from childhood into adulthood across linked census waves. Formal citizenship conferred no measurable advantage: the disadvantage operated through broken intergenerational transmission rather than levels, with father-to-son occupational-status transmission far weaker among treaty families than among white natives. Presented as a poster, IPUMS Data-Intensive Research Conference 2026, Minneapolis.

**Canales, D.A. and Enríquez, M. How Much Economic Dynamics Does a Business Register See? Auditing Mexico's DENUE, with an Application to the 2017 Puebla–Morelos Earthquake — *in preparation for Stata Journal; accepted, UK Stata Conference 2026***

Audits the reliability of Mexico's DENUE business register as a longitudinal panel, documenting entry and exit undercounts and edition-to-edition inconsistencies, and introduces the accompanying *regaudit* Stata

command for this class of diagnostic. The 2017 Puebla–Morelos earthquake serves as a demonstration case showing how naive use of the register would have produced a spurious result.

**Canales, D.A. and Enríquez, M. Damage or Dollars? Reconstruction Booms After the 2017 Puebla–Morelos Earthquake — *in preparation for Economics of Disasters and Climate Change; accepted, ICUH 2026***

Continuous-treatment difference-in-differences design using ShakeMap intensity to estimate the earthquake's effect on local establishment counts, with heterogeneity by pre-existing economic complexity. Finds a reconstruction-driven boom in establishment counts rather than net destruction, with implications for post-disaster access to essential commercial services.

## AWARDS AND SCHOLARSHIPS

- **Teaching in Excellence Award** — ITESM Campus Puebla, 2024
- **Educational Innovation Award** — ITESM Campus Puebla, 2024
- **Teaching in Excellence Award** — University of Nottingham (2017/18, 2018/19, 2020/21)
- **Graduate Teaching Fellowship** — University of Nottingham, 2018
- **CONACYT Doctoral/Masters Scholarship** — Mexican Government (2012–2014, 2015–2019)

## CONFERENCE PRESENTATIONS, POSTERS, AND WORKSHOP ATTENDANCE/DELIVERY

**UK Stata Conference (accepted)** **September 2026**
London, United Kingdom

**International Conference on Urban Health, ICUH (accepted)** **2026**

**IPUMS Data-Intensive Research Conference — poster** **July 2026**
Minneapolis, United States

**International Congress on Obesity Research, IOR — poster** **May 2026**
Monterrey, Mexico

**Junior Research Seminar** **October 2023**
Puebla, Mexico

**XI Congreso Internacional de Ciencia Política** **October 2023**
Guadalajara, Mexico

**Coloquio Mexicano de Economía Matemática y Econometría** **September 2023**
Guanajuato, Mexico

## SKILLS & QUALIFICATIONS

- Fellow of the Higher Education Academy.
- Preparing to Teach in Higher Education Certificate.
- Programming languages: R (Intermediate), Stata (Intermediate) and LaTeX (Advanced).
- Fluent in English and Spanish.